Traces of Joy: Exploring Urban Happiness through Machine Vision and Human Feeling

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MUSA 650 — Final Project Proposal

1. Problem Definition and Use Case

Where in a city will people feel happy? Conventional planning practice relies on measurable indicators such as walkability scores, traffic volumes, and property values, but rarely addresses the emotional dimension of place.

This project is adapted from a participatory study conducted by the Happiness Lab at Drexel University (Zillmer, 2025). In that study, 243 undergraduate students each identified one location in Philadelphia where they felt genuinely happy. This produced a set of 28 representative happy places, ranging from public squares such as Rittenhouse Square and cultural landmarks such as the Philadelphia Museum of Art to intimate community spaces such as the cat park in Mantua. These are not tourist destinations but ordinary places where positive emotional experiences arise in daily life.

The target users of this project are city planners and community organizations. Their core need is to identify which street-level environments are most strongly associated with happiness, thereby informing decisions around mixed-use zoning, public space investment, and streetscape improvement. The model produces a happiness probability score for every street-level sampling point across Philadelphia, forming a continuous spatial prediction surface that shows how the visual form of the city relates to subjective well-being.

2. Technical Justification

2.1 What the Model Is Learning

The model learns to identify patterns that resemble known happy places within a 14-dimensional feature space. Inputs consist of six visual features extracted from GSV panoramas through semantic segmentation: sky ratio, green view index, building ratio, road ratio, vehicle ratio, and person ratio. These are combined with eight socioeconomic variables from the 2022 ACS 5-year estimates. Critically, the model does not detect happiness directly. Predictions represent

environmental similarity to confirmed happy places, not a measurement of happiness itself. This is consistent with Naik et al. (2017), who treat computer vision outputs as correlates of urban conditions rather than causal explanations.

2.2 Why This Task Type Is Appropriate

This study faces a classic Positive-Unlabeled learning problem, known as PU Learning: we know where people felt happy, but we have no verified "unhappy" locations. Following a 200-meter sampling interval, the Philadelphia road network is expected to yield approximately 30,000 sampling points, all of which are unlabeled. Treating all of them as negative samples would introduce severe labeling bias, since many may well be undiscovered happy places.

The PU Learning framework addresses this directly. It first computes the centroid of the 28 happy places in feature space, then selects the 30% of unlabeled points farthest from that centroid as reliable negatives, and finally trains classifiers on the resulting pseudo-labeled dataset. Standard binary classification would penalize unsampled locations that happen to resemble happy places. Regression is also unsuitable, as it requires continuous happiness scores that the data does not provide.

2.3 Potential Failure Modes

The most fundamental limitation is statistical instability from the small sample size. With only 28 positive examples, each cross-validation fold contains just five to six validation positives, producing stepped ROC curves and wide confidence intervals. The happy places were also identified predominantly by university students, and most are concentrated in Center City and University City. This sampling bias may reduce the model's predictive ability for locations valued by other demographic groups.

A second concern is the hidden garden problem: GSV captures only public road perspectives and cannot photograph enclosed courtyards or rooftop spaces. Several happy places — including Cira Green rooftop park and John F. Collings Park — are exactly this type of location, meaning the visual features assigned to these points may bear little resemblance to the actual environment. Third, assigning Census Tract-level socioeconomic data to individual street points assumes spatial homogeneity within each tract, which introduces a well-known ecological fallacy. Finally, the model captures associations rather than causal relationships: dense urban blocks may concentrate

social opportunities and walkable amenities rather than being inherently happiness-inducing, and the model cannot distinguish between these explanations.

3. Methodological Precedent

3.1 Li & Ratti (2019) — Quantifying Urban Form with Google Street View

Li and Ratti show how GSV panoramas can quantify street canyon morphology at city scale. Their method transforms cylindrical panoramas into hemispherical images and applies Otsu's thresholding to estimate Sky View Factor. Sampling points are generated every 100 meters along street centerlines and results are validated against solar radiation simulations. Key limitations include restricted coverage to public road environments and substantial seasonal variation in SVF. This study adopts the same regular-interval sampling strategy, uses sky ratio as a proxy for street openness, and applies Otsu's thresholding in the baseline model.

3.2 Li et al. (2015) — The Green View Index

Li et al. propose the Green View Index, measuring street-level greenery as the proportion of vegetation pixels in GSV images, validated across multiple cities. Key limitations include high sensitivity to seasonal variation and failure to distinguish tree canopy from ground-level vegetation. The green view index feature in this study is derived directly from this method and inherits these limitations, as Philadelphia Street View imagery spans multiple collection years.

3.3 Naik et al. (2017) — Computer Vision and Urban Change

Naik et al. use time-series GSV imagery to measure changes in street appearances across five U.S. cities and link them to socioeconomic characteristics through spatially matched census data, evaluated using Kendall’s tau. The study's main limitations are the subjectivity of crowdsourced perceptual ratings and limited cross-city transferability. Their core finding, that street-level visual features can predict urban social conditions, provides the methodological foundation for this project.

3.4 Xie et al. (2021) — SegFormer

Xie et al. introduce SegFormer, a Transformer-based semantic segmentation architecture achieving strong performance on ADE20K and Cityscapes. Its hierarchical encoder without positional encoding adapts well to varying input resolutions, and the lightweight B0 variant

balances accuracy with computational efficiency. The model may underperform on scenes outside its training distribution, such as city-specific architectural styles. This study applies the ADE20K-pretrained SegFormer-B0 to all GSV panoramas via GPU-accelerated batch processing.

4. Data Plan

This study integrates four data sources, summarized in Table 1.

Data Type	Description	Source	Spatial Resolution / Coverage
GSV Panoramas	Four-directional panoramas at 3,328 x 1,664 pixels; collected across 2022 to 2024 with some seasonal variation	Google Maps Street View API	City-wide; one point every 200 m along road network; estimated ~30,000 points
Happy Places	28 student-identified happy locations with coordinates and place descriptions	Drexel University Happiness Lab (Zillmer, 2025)	Point features within Philadelphia
Road Centerlines	Street centerlines and road classifications for all accessible streets in Philadelphia, used for sampling point generation	OpenDataPhilly	Vector line features; NAD 1983 / PA South (ft)
Census Data	2022 ACS 5-year estimates; 384 census tracts; 8 socioeconomic variables including income, poverty rate, education, race, age, unemployment, and homeownership	U.S. Census Bureau API	Census Tract level; approximately 4,000 residents per tract on average

Table 1. Overview of Data Sources

The GSV API imposes daily call quotas; we will implement multi-process parallel downloading with resume-on-failure support. Sampling points lacking Street View coverage will be excluded and documented. If quotas prove insufficient, Mapillary open-source imagery serves as a fallback. Census data and the 28 happy places present no availability risk.

5. Modeling Approach

5.1 Preprocessing and Feature Engineering

Following semantic segmentation with SegFormer-B0, the 150 ADE20K categories are consolidated into six urban feature dimensions: sky, vegetation, buildings, roads, vehicles, and pedestrians. The pixel proportion of each dimension is computed as a visual feature. Census data

are spatially joined to the nearest sampling point. The eight socioeconomic variables and six visual features are then combined into a 14-dimensional feature vector and standardized using StandardScaler prior to modeling.

5.2 Baseline Model

As a performance floor, we apply Otsu's thresholding to the green view index: points above the optimal threshold are predicted as happy, those below as non-happy. This single-feature, non-machine-learning approach is analogous to NDWI thresholding in remote sensing and requires no labeled data. Greenery is among the most consistently cited positive environmental indicators in urban perception research (Li et al., 2015). We expect this single-feature baseline to perform near chance level, which would provide a clear justification for the multi-feature PU Learning approach.

5.3 Primary Model: PU Learning Framework

The primary model follows a two-stage PU Learning strategy. In the first stage, the centroid of the 28 happy places is computed in the standardized feature space. The 30% of unlabeled points with the greatest Euclidean distance from that centroid are selected as reliable negatives. In the second stage, two classifiers are trained on the resulting pseudo-labeled dataset: Logistic Regression, prioritized for interpretability, and Random Forest, prioritized for predictive performance. Both models apply balanced class weights and are evaluated using stratified five-fold cross-validation. The trained model then assigns a happiness probability score to every sampling point across Philadelphia.

This approach is preferred over end-to-end deep learning for three reasons: (1) 28 positive samples are insufficient for deep learning training; (2) the two-stage pipeline is more robust under extreme label scarcity; and (3) logistic regression coefficients directly quantify each feature's direction and magnitude, offering policy-relevant interpretability for planning decisions.

6. Evaluation Strategy

6.1 Metrics

The primary evaluation metric is ROC-AUC. Under extreme class imbalance, accuracy-based metrics are severely distorted, whereas AUC is robust to such distributions and does not depend on a specific classification threshold. It also enables direct comparison between the Otsu baseline

and the primary model. Supplementary metrics include the Precision-Recall curve and significance tests on logistic regression coefficients for interpreting the directional influence of each feature.

6.2 Data Splitting

With only 28 positive samples, a standard train/test split would leave too few positives in each subset for meaningful evaluation. This study therefore adopts stratified five-fold cross-validation, with each fold containing approximately 22 to 23 training positives and 5 to 6 for validation. AUC mean and standard deviation are reported across all folds, and results for both classifiers are compared to assess whether logistic regression's interpretability advantage comes at a cost to predictive performance.

6.3 Defining Acceptable Performance

We define two practical thresholds for planning utility. First, the model AUC must substantially exceed the Otsu baseline, demonstrating the value of integrating multiple features. Second, logistic regression coefficients must align directionally with planning theory: greenery and openness as positive, high traffic as negative. Since the model need only provide a reliable spatial ranking rather than precise probabilities, an AUC above 0.85 is considered sufficient for planning decision support. If this threshold is not met, we will consider expanding the reliable negative proportion or introducing spatial cross-validation to account for geographic autocorrelation.

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